Cornerstone Project Proposal Assignment

Section 1: Statement of the Problem (The Whys)

Minimum Words: 250

Project Title: Navigator AI

In this section, you are required to articulate the problem your Cornerstone Project seeks to address. The problem statement should be framed within a local (KLS) or broader social, cultural, or academic context, highlighting its relevance and urgency. You should explore the “why” behind their project—why this problem matters and why it needs addressing. This includes identifying existing knowledge, practice, or resource gaps and connecting the problem to the broader community or field.

In this section, you should address the following points:

- **Background Context:** Provide a brief overview of the issue, including any historical, social, or academic aspects that contribute to the problem.
- **Significance:** Explain why this problem is significant. Why should others care? Consider the impact on communities, industries, or specific groups of people.
- **Problem Statement:** Clearly define the specific problem your project will tackle. Make sure the statement is precise, focused, and feasible within the scope of your project. Please underline your problem statement.

In most schools, essential information — like policies, schedules, and assignments — is scattered across emails, PDFs, and learning platforms. Students waste time looking for answers, parents miss important updates, and teachers spend hours re-explaining the same details instead of focusing on instruction. As schools adopt digital tools such as Canvas and Google Drive, the lack of a unified and intelligent system for managing information has become a major barrier to communication and learning.

At KLS, where students are encouraged to lead their own education, clear access to accurate information is critical for self-directed learning. But the current system often forces students to dig through multiple platforms to find due dates or mastery goals, reducing autonomy and focus. Teachers face similar challenges when trying to identify patterns in student progress or plan lessons efficiently. Parents, too, struggle to stay informed without a central source of updates.

The problem this project addresses is the absence of a secure, intelligent platform that organizes school knowledge, personalizes learning insights, and assists teachers and families in navigating academic life more efficiently.

This issue matters because it impacts how the entire community learns and collaborates. By combining document organization, personalized student dashboards, curriculum insights for teachers, and clear communication for parents, an AI-powered platform can make school life more streamlined, transparent, and supportive for everyone involved.

Section 2: Process (Materials, Preparations, and Steps)

Minimum Words: 350

This section should detail the methodology and steps that will be taken to address the problem outlined in Section 1. You are expected to provide a comprehensive plan of action that demonstrates critical thinking and careful planning. The process should outline the steps and identify key materials and resources required, including who or what groups will be impacted or involved.

In this section, you should cover the following elements:

- **Materials and Preparations:** List and describe all materials and resources needed for the project, such as tools, equipment, software, or (other) resources. Explain any preparatory steps necessary to ensure the project's success (e.g., obtaining permissions, conducting initial research, or gathering data).
- **Key Stakeholders and Impact:** Identify who will be impacted by the project. Who are the primary beneficiaries? Are there partners, organizations, or community members involved? Explain the role of each participant. Again, a Cornerstone project should impact others (locally, globally, or *glocally*[1])
- **Step-by-Step Plan:** Detail at least <u>two</u> to <u>three</u> major steps required to achieve the project's objectives. Each step should briefly explain the actions involved and any expected outcomes. For example, Step 1 might involve conducting surveys or interviews, while Step 2 focuses on analysis and design.

Materials and Preparations

- Software Tools: Python, OpenAI API or similar LLMs, a vector database (Pinecone or FAISS), and a frontend framework (React or Streamlit).
- Data Sources: Real KLS documents (schedules, handbooks, and policies) with staff permission.
- Hosting: Secure deployment using Google Cloud or Vercel.
- Testing Participants: Students, teachers, and administrators for usability feedback.
- Research and Permissions: Collaborate with KLS staff to securely use official school documents while maintaining privacy and accuracy.

Key Stakeholders and Impact

- **Students** benefit from personalized and quick answers about schedules, assignments, or policies, fostering independence and time management.

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[1] The *glocal* is a mixed concept of globally shared norms, attitudes, behavior, and actions, that allow individuals as well as groups and organizations to take decisions and resolve issues locally (Mihr, 2022).

- **Teachers** gain an intelligent assistant that can handle repetitive queries and give insights into class trends, helping refine curriculum and pacing.
- **Parents** access a single, trustworthy platform for all school updates, making communication clearer and more consistent.
- **School Administrators** gain a tool that reduces communication bottlenecks and improves transparency.

Step-by-Step Plan

Step 1: Research and Data Collection
Work with school staff to identify key documents and data sources. Collect and organize materials, then clean and format them for AI processing. This step also includes interviewing teachers and students to understand the most common information gaps and user needs.
Expected Outcome: A structured dataset and a clear understanding of user priorities.

Step 2: Build and Train the AI System
Develop a backend that processes and indexes documents into a searchable database. Connect it to a language model that can retrieve and summarize information accurately. Integrate Canvas to enable personalized features like due-date summaries and learning outcome tracking.
Expected Outcome: A functioning knowledge engine that can answer questions and generate personalized insights.

Step 3: Design and Test the User Interface
Create a user-friendly web interface that includes search, chat, and personalized dashboards for students and teachers. Conduct multiple testing rounds with volunteers from each user group, collecting feedback on clarity, accuracy, and usability.
Expected Outcome: A polished + accessible prototype ready for broader school testing.

Step 4: Evaluate and Refine

Measure how much faster and easier users can find information compared to existing methods. Collect survey data on satisfaction and accuracy, and make final improvements based on feedback.
Expected Outcome: A reliable + school-ready platform that simplifies communication and supports learning for the entire KLS community.

Section 3: Goals (Final Product)

Minimum Words: 200

The proposal's final section should describe the project's desired outcomes and end product. This is the opportunity for you to visualize the impact of your work and define success in concrete terms. The goals should be ambitious yet realistic and measurable to evaluate the project's effectiveness upon completion.

In this section, you should address the following:

- **Final Product Description:** Describe the anticipated end product, whether it is a report, a prototype, an event, an artwork, or another form of deliverable. Be specific about the final product's format, scope, and intended function.
- **Intended Impact:** Explain how the final product will address the problem identified in Section 1. What change, improvement, or knowledge contribution does the project aim to achieve? Please underline the intended impact.
- **Measurable Success Criteria:** Define at least <u>two criteria</u> by which the project's success will be measured. This could include qualitative feedback, quantitative data, project reach, or impact assessments.

Final Product Description

The final product will be an AI-based school platform that brings together all essential information — policies, schedules, announcements, and course resources — into one organized, searchable space. It will feature a conversational chatbot that can provide quick, source-based answers, while administrators control editing access to ensure information stays accurate and secure.

For students, Navigator AI will offer personalized academic guidance, drawing data directly from Canvas to display upcoming work, track learning progress, and highlight areas that need attention. Teachers will gain an assistant that helps analyze class performance, suggest pacing changes, and reduce time spent on repeated logistical questions. Parents will finally have a clear, dependable place to check school updates and key documents. Together, these tools create a smoother, more connected learning environment.

Intended Impact

The intended impact is to create a clear, efficient, and personalized digital infrastructure that helps schools communicate better, supports student independence, and lightens the workload for teachers and administrators.

Measurable Success Criteria

- Accessibility and Efficiency: At least a 50% decrease in the time required to locate school information.
- User Experience: 85% or higher satisfaction ratings from students, teachers, and parents.
- Learning Support: Demonstrated improvement in student organization and mastery tracking.
- Reliability: 95% verified accuracy in responses and strong data-privacy safeguards.